

Bilkent University, CS 473/573, Session 4, 32pts, 40min

Q4: You are to organize a canoe trip down the river. Canoes are available for rent at a sequence of trading posts along the river. You will start your trip by renting a canoe at post 1 (where the canoeing begins) and end the trip at post n (where the canoeing ends). You are allowed to stop at any post, drop off the canoe you have and rent another one. For all pairs (i, j) with $i < j \leq n$, the cost of renting a canoe at post i and dropping it off at post j is provided. For example, if there are four posts in total, the costs might be:

$R[i, j]$	$j = 1$	2	3	4
$i = 1$	-	10	15	50
2	-	-	40	20
3	-	-	-	35
4	-	-	-	-

Your job is to design an **efficient** algorithm to plan the sequence of canoe rentals to **minimize the total cost**. In the example shown, an optimal solution is to rent a canoe from post 1 to post 2 (cost 10), then rent a canoe from post 2 to post 4 (cost 20) with a total cost of 30. Your grade will be based on the efficiency of your algorithm.